YESHI DOLMA

Seattle, WA | yeshidpn@gmail.com | linkedin.com/in/dolmayeshi

PROFILE

Applied machine learning scientist and health-data researcher with 9+ years of experience in predictive modeling, causal evaluation and large-scale data systems, now focused on care transitions and complex care. Founder of an early-stage venture on post-discharge recovery for patients and family caregivers. Recent work evaluates AI-enabled global health programs and Seattle's alternative 911 crisis response. Skilled at translating technical findings into decisions for leadership, program teams and communities.

Focus: Care Transitions & Family Caregiving; Program Evaluation & Impact Measurement; Causal & Predictive Modeling; Real-World Data; Geospatial & Spatiotemporal Analysis; Bias-Aware Causal Inference; Data Quality & Integrity; Health Equity; Responsible AI;

RELEVANT TECHNICAL SKILLS

Applied AI & Product Development: Python, SQL, NLP, LLMs, retrieval-augmented generation (RAG), embedding models, predictive modeling, prototype development, model evaluation.
ML & Advanced Modeling: Predictive modeling, DNNs, transformers, GBDTs, NLP, embedding models, LLMs, RAG workflows, spatiotemporal analysis, observational analysis, model evaluation
Public Health & Equity Analytics: Statistical modeling, causal inference, observational analysis, cohort definition, biostatistics, spatiotemporal analysis, behavioral health and emergency-response data, social determinants of health, equity measurement.
Patient & Caregiver Discovery: Discovery interviews, unmet-needs assessment, patient and caregiver journey mapping, clinical and operational workflow analysis, translation of findings into product requirements.
Evaluation & Responsible AI: Outcome and metric design, A/B testing, clinical AI evaluation, bias and drift monitoring, data validation, reproducible analytics, auditability.
Data Systems & Deployment: Microsoft Azure, AzureML, large-scale data pipelines, distributed processing, MLOps, production monitoring, Git/GitHub.
Cross-Functional Leadership: Technical project scoping, stakeholder engagement, collaboration with clinical and community partners, communication of technical findings to nontechnical audiences

EXPERIENCE

Founder | Early-Stage Venture: Post-Discharge Recovery & Care Transitions
Aug 2026 – Present
Seattle, WA

- Developing accessible, voice-first support to help patients and family caregivers understand care plans, manage daily responsibilities, and communicate needs between visits and after discharge.
- **Discovery research:** Conducting **75+ patient and caregiver conversations,** complemented by interviews with advocates, community health workers, clinicians, and health-system leaders, to identify unmet needs across complex conditions, including gastroparesis.

Wadhwani AI | Technical Consultant, AI Evaluation & Impact Measurement
Apr 2026 – Jul 2026
Global Health | Contract | WIAI US

- **Evidence-Based Impact Frameworks:** Developed analytics reporting and evaluation frameworks for AI-enabled healthcare programs across diabetic retinopathy, anthropometry, dermatology, and oral cancer screening, translating clinical, operational, and technical goals into measurable indicators.
- **Clinical Data Quality & Validation:** Defined metrics for data completeness, input quality, subgroup performance, workflow feasibility, and model readiness to support real-world deployment decisions.
- **Technical Translation:** Translated grant objectives, technical milestones, and program-level goals into measurable reporting indicators, acting as liaison between engineering teams, program teams, and non-technical leadership.
- **Data Quality & Deployment Risk:** Assessed risks that could affect real-world use, including poor input quality, model bias, limited generalizability, and integration challenges in heterogeneous, resource-constrained healthcare settings.
- **Workflow & Implementation Risk Assessment:** Identified risks affecting real-world healthcare adoption, including poor input quality, model bias, limited generalizability, referral workflow gaps, and inconsistent deployment settings.
- **Continuous Improvement Metrics:** Designed monitoring indicators to evaluate whether AI solutions remained clinically useful, operationally feasible, and adaptable across changing healthcare contexts.

Microsoft Corporation | Applied ML Scientist (Tech Lead)
Aug 2016 – Dec 2025

Data Products & Enterprise Analytics

- **Data Products:** Created standardized feature pipelines and reusable datasets with documented transformation logic to support reliable downstream analytics.
- **Analytical Solution Development:** Designed and maintained large-scale analytical data products used to evaluate system performance, product behavior, and business outcomes across global production environments.

- **Data Integrity & Monitoring:** Created automated data quality checks, drift monitoring, and performance diagnostics to identify anomalies, regressions, and data stream failures before they affected decision-making.
- **Resource & Deployment Planning:** Scoped pipeline deployments, project development, managed compute budgets, and allocated engineering resources for large-scale systems.
- **Executive & Stakeholder Communication:** Translated complex statistical findings and system behavior into clear recommendations for product, engineering, and leadership stakeholders.

Statistical Evaluation & Experimentation

- **Trend & Impact Analysis:** Used statistical analysis, A/B testing, matching methods, and quasi-experimental approaches to compare outcomes, identify trends, and isolate true incremental impact.
- **Bias & Confounding Mitigation:** Built analytical corrections for selection bias, confounding, and covariate shifts to improve reliability of performance comparisons across diverse user segments and markets.
- **Operational Diagnostics:** Developed monitoring workflows and dashboards to detect system regressions, segment-level performance issues, and workflow inefficiencies.
- **Mentorship & Leadership:** Mentored junior developers and interns and served as technical lead for complex analytics, modelling and deployment projects.

Predictive Modeling & ML Engineering

- **Full Lifecycle Ownership:** Led the development of production ML systems from data ingestion and feature engineering through to online serving and monitoring.
- **Large-Scale Predictive Modeling:** Built scalable predictive modeling pipelines (DNNs, transformers, GBDTs) to extract insights from sparse behavioral datasets for user engagement prediction at global scale.
- **Distributed Cloud Data Frameworks:** Trained terabyte-scale models using distributed cloud computing frameworks, consolidating hundreds of independent pipelines into unified, reusable modeling infrastructure.
- **Efficiency Engineering:** Reduced storage and compute cost by 4–10% via algorithmic feature pruning and model compression techniques while maintaining predictive performance constraints.

SELECTED PUBLIC HEALTH & POLICY RESEARCH

Alternative 911 Emergency Response: Seattle CARE Evaluation, Equity & Service Gap Analysis

Independent Researcher (In collaboration with Seattle's CARE Department) *2026*

- **Administrative Data Analysis:** Integrated 911 dispatch records, census data, and socioeconomic indicators to evaluate service gaps, crisis response patterns, and opportunities for alternate emergency response deployment.
- **Metric & Cohort Design:** Developed reproducible definitions for CARE-eligible call groups, behavioral health crisis indicators and safety exclusions. Designed metrics to map neighborhood-level service gaps.
- **Crisis Triage Framework:** Modeled the population-level incidence of behavioral health crises, identifying 6.4% of 911 events as behavioral health crises and ~45% qualified under broader Tier-1/Tier-2 baselines to provide a data-driven baseline for expanding non-police interventions.
- **Equity-Driven Analytics:** Analyzed socioeconomic determinants of crisis burden using multivariate regression and RSE indices; found high-vulnerability zones experience 3× higher crisis rates, significantly correlated with neighborhood-level socioeconomic factors ($\rho \approx 0.44$).
- **Resource Allocation & Deployment Modeling:** Designed a place-based CARE deployment strategy that balances equity and efficiency through residential and activity-driven placement to optimize resource-constrained environments.
- **Executive-Ready Reporting:** Produced decision-support analyses to quantify demand, identify high-need geographies, and support strategic planning for non-police crisis response expansion.
- **Data Quality & Bias Assessment:** Evaluated limitations in administrative reporting, missingness, and incident classification to improve interpretability and support responsible use of operational data.
- **APHA 2026 Oral Presentation:** Selected for an oral presentation at the American Public Health Association (APHA) Annual Meeting 2026 co-authored with Seattle's CARE department (City of Seattle).

OTCSafe: Agentic AI for Pharmacological Safety & Clinical Decision Support *2026*

- **Multi-Source Data Integration:** Built a multi-stage data pipeline converting free-text inputs into risk reports by integrating NLP extraction with secondary pharmacological databases (RxNorm/OpenFDA) via the Claude API.
- **Medication Safety Engine:** Programmed logic to resolve brand-to-ingredient mappings, detect drug interactions, and aggregate cumulative dosages to surface high-risk polypharmacy patterns.
- **Data Quality & Verification:** Developed hallucination guards and cross-step verification evaluated against real-world queries to ensure source-grounded, reproducible analytical outputs.
- **System Benchmarking:** Evaluated and benchmarked performance metrics, demonstrating the superior reliability of tool-augmented agentic workflows over single-pass LLMs; delivered as a prototype.

COVID-19 Misinformation Dynamics & Policy Impacts *2025*

- **Causal Analysis Pipeline:** Modeled Google Trends as a proxy for misinformation engagement, correlating composite indices with longitudinal COVID-19 health trajectories in Washington State.
- **Signal Processing Workflows:** Engineered workflows for data smoothing, normalization, and policy-event annotation to accurately isolate drivers of public misinformation surges
- **Policy Mechanism Identification:** Discovered that engagement was driven by policy communication vacuums and guidance reversals rather than disease severity, revealing that institutional skepticism persists from cumulative credibility erosion.
- **Data-Driven Policy Insights:** Demonstrated that public health interventions must prioritize transparent, structural institutional communication over traditional models that focus solely on addressing information gaps.
- **Analytical Extensions:** Scaling the framework using NLP topic modeling and sentiment analysis to track multi-source narrative evolution and cross-state policy variation.

EDUCATION

Integrated Dual Degree (B.Tech. & M.Tech.), Computer Science & Engineering **2011–2016**
Indian Institute of Technology, Kanpur (IIT Kanpur) GPA: 8.3/10
Academic Minor: English Literature GPA: 10/10

- **Master's Thesis: Zero-Shot Learning of Image Categories using Gaussian Processes**
  - published at **ACCV 2016** (tier-1 conference) and Springer 2017;
  - presented at **Grace Hopper Conference India** 2017.

Teaching Assistant

- *Computer Science and Engineering Communication Skills* | Fall 2015
- *Machine Learning for Computer Vision* | Spring 2016

Continuing education: relevant coursework in Public Health -
Johns Hopkins University (Online): Infectious Disease Transmission Models; Essential Epidemiologic Tools; Data and Health Indicators in Public Health; Foundations of Health Equity Research; Design and Conduct of Clinical Trials; (*ongoing Harvardx-)* Prescription Drug Regulation, Cost, and Access: Current Controversies in Context*;*
IHME: Introduction to the Global Burden of Disease

PUBLICATIONS

- **Dolma, Y.**, Barden, A., & Trejo, M. (2026). *Alternative 911 emergency response: Measuring service gaps, equity, and deployment opportunities in Seattle's CARE model.* Conference acceptance at American Public Health Association Annual Meeting (APHA).
- **Dolma, Y.,** Kalani, R., Agrawal, A., & Basu, S. (2021). Improving Bounce Rate Prediction for Rare Queries by Leveraging Landing Page Signals. WWW Workshop on Transforming Big Data into Actionable Knowledge (BiDAW), Ljubljana, Slovenia.
- **Dolma, Y.**, & Namboodiri, V. P. (2016). Using Gaussian Processes to Improve Zero-Shot Learning with Relative Attributes. Asian Conference on Computer Vision (ACCV), Taipei, Taiwan.

SELECTED AWARDS

- **Greatness Award - One Microsoft, 2021:** Recognized for cross-team collaboration and global product delivery.
- **Greatness Award - Engineering Excellence, Microsoft Advertising, 2020.**
- **Recognition of Contribution Award, Microsoft Advertising, 2017.**
- **Top Design Idea, Microsoft Code.Fun.Do, 2012**
- **2nd Prize, GE Healthcare Hackathon, 2017**
- **Women Emerging in Finance Event, Goldman Sachs, 2015.**
- **AISSCE State Topper, 2010:** Ranked first across all schools in the state with 95.2%.
- **NTSE Scholar, 2008:** National-level merit scholarship by the National Council of Educational Research and Training, India.
- **Nationwide Interactive Math Olympiad, 2008.**

VOLUNTEER EXPERIENCE

- **Juneteenth Career Support Volunteer** - career-support efforts by reviewing resumes and mentoring individuals interested in entering data science roles.
- **Tibetan Association of Washington Volunteer** - Supported Tibetan diaspora community engagement through event planning, design ideas, and coordination of cultural and community activities.